Paper 10 - T10 Master Receipt

CQE-paper-10
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

Bind Paper 00 and Papers 01-09 into a replayable master receipt carried from the observer's step-00 enumeration event into step 1, with typed transport rows, materialized lookup receipts, and visible open-lift boundaries.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-10.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-10\paper_kernel_manifest.json
- body: production\papers\CQE-paper-10\PAPER-BODY.md
- source: production\papers\CQE-paper-10\SOURCE.md
- formal: production\papers\CQE-paper-10\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-10\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-10\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-10.md

Paper 10 - T10 Master Receipt

Abstract

Paper 10 proves the first stack-level receipt theorem in the CQECMPLX suite. Its object is not a new physical mechanism. Its object is the proof that Paper 00 and Papers 01-09 are bound into one inspectable and replayable unit.

The master receipt verifies:

```
Paper 00 as inherited information-burden contract
Paper 00 as observer enumeration center C00
the 00 -> 1 encoding event
Papers 01-09 as pass-like formal receipt bindings
typed transport rows
local witness replay
lookup cache materialization
visible open lifts
```

The theorem is closed for receipt integrity. It does not claim that every registered lift is already demonstrated. Its honest verdict is a passing master receipt with open-lift boundaries still visible.

Claims

Claim 10.1. Paper 00 and Papers 01-09 can be bound into one replayable T10 master receipt.

Claim 10.2. Papers 01-09 have promoted formal receipts with pass-like status.

Claim 10.3. The transport table has four typed rows, valid classifications, replaying local witnesses, two demonstrated rows, and two visible non-demonstrated lifts.

Claim 10.4. The lookup cache materializes the required Rule 30, unipotent orbit, lattice form, UMRK, and LMFDB registers.

Claim 10.5. The Prize 3 lookup substrate keeps `closed_form_claim = false` and therefore does not overclaim cold-start closure.

Definitions

A **paper receipt binding** is a pair:

```
(paper_id, receipt_path)
```

where the receipt exists, can be parsed, and reports a pass-like status for the theorem it carries.

The **observer center `C00`** is the active center introduced by the requested enumeration event at Paper 00. It is the encoded fact that an observer asked the system to enumerate something.

The **`00 -> 1` encoding event** is the transition from the inherited Paper 00 burden contract into the first active carried state in Paper 1.

A **transport obligation row** is:

```
id
source_object
target_object
map
```

```
preserved_quantity
failure_condition
witness
classification
proof_boundary
```

Allowed classifications are:

```
demonstrated
bounded_local
bounded_external
registered_landing_forms
open
```

A **lookup receipt** is:

```
kind
key
value
source_id
evidence_level
complexity_claim
```

A **T10 master receipt** is:

```
T10 = (C00, E00->1, P00, P01..P09, R, L, V, O)
```

where `R` is the transport table, `L` is the lookup cache, `V` is the verifier verdict, and `O` is the visible open-lift set.

Theorem 10.1 - T10 Master Receipt Integrity

The CQECMPLX substack consisting of Paper 00 and Papers 01-09 is inspectable and replayable as a single receipt object. The receipt proves the existence and integrity of the bindings, transport rows, local witnesses, lookup receipts, and visible open-lift boundaries.

Proof

First bind Paper 00 by requiring its source contract and at least one Paper 00 proof receipt. This establishes Paper 00 as the inherited burden contract. It also establishes `C00`, the observer's requested enumeration encoded as the system center.

The transition from Paper 00 to Paper 1 is therefore an active encoding event:

```
observer request -> C00 -> E00->1 -> first carried paper state
```

For each paper from 1 through 9, bind the paper to its promoted formal receipt. Each receipt must exist and report a pass-like status. These bindings form the paper component of `T10`.

Next construct the transport table using the four registered rows:

```
LCR_TO_D4_AXIS_SHEET
D4_TO_J3O_DIAGONAL_CARRIER
J3O_TO_G2_F4_T5A_ROUTE
EXCEPTIONAL_ROUTE_TO_NIEMEIER_LANDING_FORMS
```

The verifier checks required fields and valid classifications. It replays the local witnesses:

```
verify_chart_codec_d4
verify_j3o_axioms
verify_conjugate_triple
```

```
verify_niemeier_landing_registry
```

The transport verdict is:

```
pass_with_open_lifts
```

with two demonstrated rows and two visible non-demonstrated lifts. This proves inspectability, not closure of all lifts.

Finally, materialize the lookup cache. The cache exposes:

```
rule30_bits = 1000000
unipotent_orbits = 157
lattice_forms = 24
source_registers.umrk = true
source_registers.lmfdb = true
```

The Prize 3 lookup receipt is accepted only because it keeps:

```
closed_form_claim = false
```

and records the remaining cold-start obligation. Therefore all components of `T10` are present, typed, replayable, and honest about their boundaries.

Receipt

The primary executable receipt is:

```
production/formal-papers/CQE-paper-10/verify_t10_master_receipt.py
production/formal-papers/CQE-paper-10/t10_master_receipt.json
```

The receipt status is `pass`. It verifies:

```
paper00_contract_bound          = true
observer_center_encoded         = true
papers01_to_09_receipts_present = true
papers01_to_09_status_pass_like = true
transport_rows_inspectable      = true
transport_witnesses_replay      = true
transport_open_lifts_are_visible = true
lookup_cache_materializes       = true
```

The transport summary is:

```
status = pass_with_open_lifts
row_count = 4
demonstrated_count = 2
open_lift_count = 2
all_lifts_demonstrated = false
```

Falsifiers

The verifier rejects:

```
T10 proves every registered lift is already demonstrated
the lookup cache makes a cold-start closed-form N-to-fingerprint claim
a paper enters the master receipt without a source or receipt binding
a later paper can ignore the observer enumeration event encoded at 00 -> 1
```

Role in the Suite

Papers 1-9 build the first carried chain after the observer enumeration event:

```
LCR carrier
correction surface
triality surface
boundary repair
path carrier
causal code
discrete-continuous bridge
lattice closure template
Hamiltonian window emergence
```

Paper 10 wraps that chain as a stack-level audit object:

```
observer request at Paper 00
-> C00
-> 00-to-1 encoding event
-> paper receipts
-> transport rows
-> local witness replay
-> lookup receipts
-> pass verdict with visible open lifts
```

Open Audit Boundaries

- Paper 10 does not close the cold-start `N`-to-axis/sheet map.
- Paper 10 does not close the `N`-to-Weyl-fingerprint map.
- Registered Niemeier landing forms are valid receipt targets, not automatic proof closure.
- Later papers may cite T10 as a master receipt, but they must still prove their own domain claims.
- Later recentering requires an explicit new observer/enumeration event and handoff.

Conclusion

Paper 10 proves that the first CQECMPLX substack is not a pile of adjacent claims. It is a single replayable master receipt carried from the observer's initial enumeration event. The result is proof-first: `C00`, `E00->1`, paper bindings, transport rows, lookup receipts, and open boundaries all appear in one verifiable object.